Variable argument list

Consider a function, which has return type and can accepts variable number of arguments etc like printf etc.....

```
int printf(const char *format, ...);
```

```
#include<stdio.h>
int main()
{
	char s1[]="123",*p="456";
	printS("hello\n");
	printS("abc","def","xyz");
	printf("\n");
	printS("A","B",s1);
	printf("\n");
	printS(s1);
	printf("\n");
	printS(p);
}
 void printS(char *p,...)
 {
	puts(p);
 }
```

//

```
 #include <stdarg.h>
 void va_start(va_list ap, last);
 type va_arg(va_list ap, type);
 void va_end(va_list ap);
 va_list;
```

va_list===>this is a macro
	====making arrangement of data
 va_start (handle,p)//mapping between calling function
Two ways.....
  printS(“hello”,Null);

//

```
#include<stdio.h>
#include<string.h>
#include<stdarg.h>
int main()
{
	char s1[]="123",*p="456";
	 printS("hello",NULL);
	 printf("\n");
	 printS("BYE","haii",NULL);
```

```
}
  void printS(char *p,...)
 {
       va_list handle;
       va_start(handle,p);
        puts(p);
        while(p=va_arg(handle,char *))
        {
             puts(p);

        }
        va_end(handle);

  }
```

//

va_list==>data type for handler to get the data

handle==new pointer
va_argv will be holding the next argument

String passing ,Null is taken as terminator

//

```
#include <stdio.h>
#include<stdarg.h>////data  canb be 0..will not work
int main()
{


    int x = 10, y = 20;
    printf("%d", sum(x, y,0));
    // printf("%d", sum(1, 21, 0, 34, 5));
    printf("%d",sum(2,x,y));
    return 0;
}

int sum(int v, ...)
{
    int s = v;
    va_list ap;
    va_start(ap, v);
    while ((v = va_arg(ap, int)))
    {
        s += v;
    }
    va_end(ap);
    return s;
}
```

//

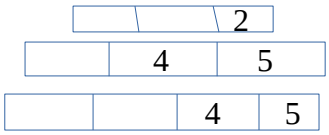
```
#include <stdio.h>
#include<stdarg.h>
int summ(int cnt, ...);

int main()
{
    int x = 10, y = 20;
    printf("%d",summ(2,x,y));
    printf("\n");
    printf("%d",summ(6,1,1,x,1,1,1));
    return 0;
}

int summ(int cnt, ...)
{
    int s = 0;
    va_list ap;
    va_start(ap,cnt);
    while(cnt--)
    {
        s+=va_arg(ap,int);
    }
   va_end(ap);
   return s;
}
```

printf

```
#include <stdio.h>
int main()
{
    printf("%3d\n",2);
    printf("%3d\n",45);
    printf("%4d",45);
}
```

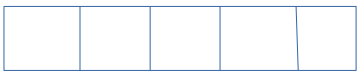



```
q:#include <stdio.h>
int main()
{
    printf("%3d\n",2);
    printf("%3d\n",45);
    printf("%4d\n",45);
    int x=5,y=6;
    printf("%*d\n",x,y);
    printf("%*d",y,x);

   printf("\n");
   int a=10,b=65;
```

```
 int n=printf("%*c",a,b);
 printf("\n%d",n);


 ///add without using +
 int k=5,v=3;
 n=printf("%*c",x,32);//////
 printf("\n%d",n);
printf("\n");
n=printf("%*c%*c",k,32,v,32);
printf("\n%d",n);//8
```



n=5